VENTING? AMONG US

Proj 07: Linear to rotary motion

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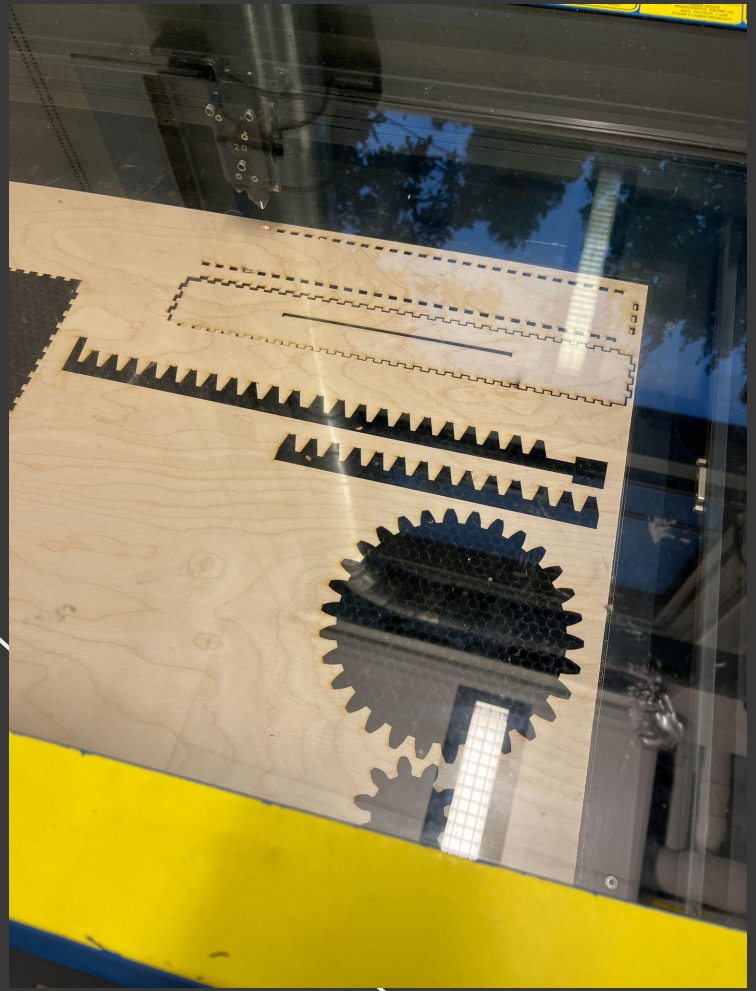
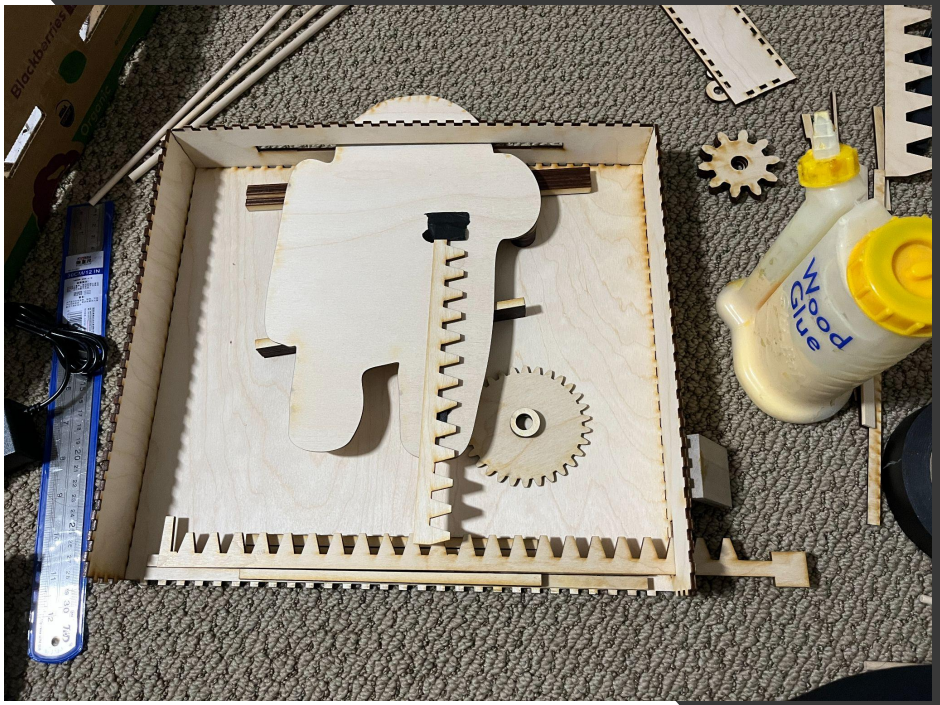
PRODUCT DESCRIPTION

An iconic and well-known action from a mobile game called “Among Us” – an adaptation of “venting”.

Video Link: https://youtu.be/P4UvP38Zs94




Wood Glue

PROCESS

The first idea that came to mind when thinking of linear motion was the main character from a popular game “venting” up and down.

After designing the initial pieces needed to set the idea into motion, figuring out the linear motion was difficult since there were more obstacles than expected. For instance, I wanted the character to be inside a box, imitating a vent, but this caused different design changes to be implemented.

The majority of the parts initially cut weren’t used because they were the wrong size and didn’t work well with the new design changes. The size of the product was also extremely large which caused hesitation since material was limited.

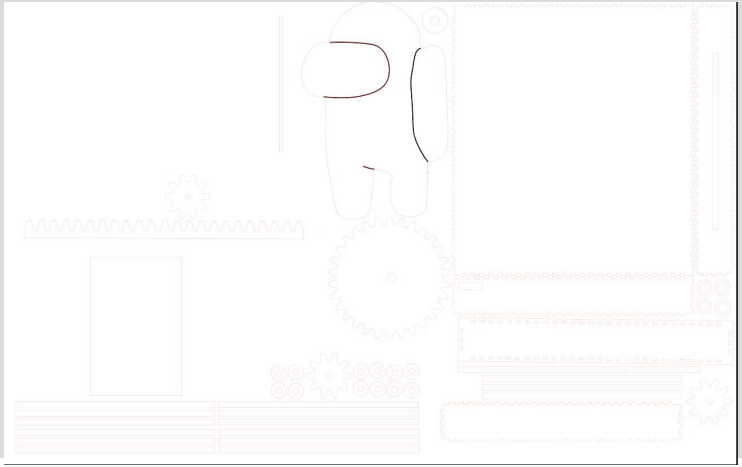
The linear motion of the character moving up and down was the biggest struggle since it was very large and the box conflicted with the actual movement desired: the character being fully inside the box then popping its head out.

After heavily experimenting with different “solutions” to the box constraint, a solution was found and everything was finally working. A lot of redesigning, experimenting and scrap material was used for this project that isn’t exactly shown by the final product.

Bill of Materials

- 1 - Wooden dowel
- Birch plywood
- Acrylic plastic
- Wood glue
- Handsaw
- Sandpaper
- Paste wax
- Laser cutter
- X-acto knife
- Black Gaffers Tape

ADOBE ILLUSTRATOR LASER CUTTING FILE



Link to Google Folder of Files:
https://drive.google.com/drive/folders/1TjRMDZSie9tJHrCpOqKivn2IKgRiMT0p?usp=sharing

Rotary-to-linear mechanism

My rotary-to-linear mechanism involved having a tooth rack that users would pull horizontally back and forth to make the Among Us cutout pop up and down. There are 4 mains components - two teeth rack and 2 gears glued together - that propelled the cutout up and down.

I added paste wax on all of the turning parts to make the motion smoother since there was a lot of initial hesitation between all of the pieces. My cutout was also not aligned in place so there is a little leeway, allowing jams to occur.

REFLECTION

SURPRISES ENCOUNTERED

I thought that this project would be fairly straightforward and simple to make but there were a lot more obstacles than expected. I had completely changed my initial design to incorporate the box element which caused significantly more work since I needed to find a way to support each of the moving parts. The size of my product was also a surprise since I didn't know it would cause this much difficulty.

FRUSTRATING PARTS

Getting the character to move up and down was the most frustrating part for this project. The multiple gears and tooth racks that I had cut out didn't fit well with each other so I had to repeat that process over and over. Additionally, there was a lot of space between the character and the rest of the box so my gears kept getting caught onto different parts, causing the gears to jam and not turn unless fixed.

DIFFERENT APPROACHES

I would have made the size of this project significantly smaller if I had the chance to redo it. That would make support and stability a lot easier to maintain. Additionally, I would try a different mechanism since I don't really like how the handle to put the character into motion is sticking out of the box — I would prefer a handle that can be easily turned. I would also make the entire box removable so people can see the inner workings.

THANK YOU!